

Loan Default Risk Analysis

End-to-End Project Report

Oct 7, 2026 · @Vanshika

Executive Summary

This project builds a complete, end-to-end credit risk analytics pipeline for retail lending: a realistic 2,000-loan synthetic dataset, a Python-based data validation and feature-engineering layer, and an interactive Power BI dashboard that surfaces where default risk concentrates across a loan portfolio.

The finished dataset spans 31 original fields (demographics, loan terms, credit bureau data, and financial ratios) across 15 Indian states, enriched in Python with six derived fields — age band, credit score band, DTI band, LTV band, loan security flag, and application year — to support filtering and segmentation in the dashboard. The portfolio carries an overall **default rate of 12.65%** (253 of 2,000 loans), and the analysis finds that risk is driven overwhelmingly by three factors: loan type (Business Loans default at 23.7%, more than double the portfolio average), employment stability (Gig/Freelance borrowers default at 27.5% versus 3.8% for Government-salaried borrowers), and credit score (Poor-band borrowers default at 45.0% versus 1.1% for Excellent-band borrowers).

The dashboard translates these findings into a single-screen, filterable risk view — built with a dark, glass-panel visual theme, a dynamic field-parameter chart that lets a user re-cut default rate by any risk factor on demand, and a dedicated Credit + Financial Risk Explorer panel — so a credit team can move from a portfolio-level number to an actionable segment in a few clicks.

1. Project Overview & Objective

Lenders approve loans before they can observe repayment behaviour, so the only lever they have at underwriting time is the applicant's profile: who they are, what they earn, what they're borrowing for, and what their credit history says about them. The business problem this project addresses is identifying, from that profile alone, which segments of a loan book are most likely to default — so a credit team can price risk correctly, tighten underwriting on weak segments, and monitor the portfolio over time.

Objective. Build an analytics pipeline that takes a loan portfolio from raw data to a decision-ready dashboard, answering four questions:

1. What does the portfolio look like today — how many loans, how much is disbursed, what's the current default rate?
2. Which loan types, employment categories, geographies, and borrower segments carry the most default risk?
3. How do credit bureau signals (score, utilization, delinquency history) and financial ratios (DTI, LTV, interest rate) relate to default?
4. Has portfolio risk been improving or worsening over time?

Scope. The project covers three stages: (1) a synthetic but behaviourally realistic loan dataset standing in for a bank's loan book, (2) a Python notebook that validates the data and engineers the segment fields the dashboard needs, and (3) a Power BI dashboard built for a credit risk or portfolio management audience.

Intended audience. Credit risk analysts, portfolio managers, and underwriting teams who need a repeatable view of where default risk sits in a lending book — and, as a portfolio project, anyone evaluating end-to-end data analytics skill from raw data through to a polished BI deliverable.

2. Dataset Description

The base dataset (`loan_default_risk_dataset.xlsx`) holds **2,000 loan records across 31 fields**, spanning 15 Indian states and seven loan products. It was generated to mirror the statistical relationships found in real Indian retail lending — income scaling with employment type, education, and city tier; EMI computed from the standard reducing-balance formula; credit score and default risk driven by a multivariate model rather than assigned at random — so that every downstream relationship in the dashboard (score vs. default, DTI vs. default, and so on) holds together the way it would in a real loan book.

Column	Description	Range / Values
Loan_ID	Unique loan identifier	LN100001 - LN102000
Application_Date	Date the loan was applied for	Jan 2022 - Dec 2025
Age	Borrower age (years)	21 - 68

Column	Description	Range / Values
Gender	Borrower gender	Male / Female / Other
Marital_Status	Marital status	Single / Married / Divorced / Widowed
Dependents	Number of financial dependents	0 - 4
Education_Level	Highest education	5 levels, school to professional degree
Employment_Type	Primary income source	6 categories (Salaried Govt/Private, Self-Employed Business/Professional, Gig, Retired)
Years_in_Employment	Years in current occupation	0.5+
Annual_Income_INR	Gross annual income (₹)	1.2L - 60L
State	State of residence	15 Indian states
City	City/town of residence	Mapped to state
City_Tier	Metro / Tier-2 / Tier-3 / Rural	Geography risk proxy
Residence_Type	Owned / Rented / Family-Parental	—
Loan_Type	Product type	Home, Personal, Auto, Education, Business, Gold, Loan Against Property
Loan_Amount_INR	Sanctioned principal (₹)	Scaled to income and product
Tenure_Months	Loan tenure	6 - 360 months
Interest_Rate_Pct	Annual interest rate	7% - 26%
EMI_INR	Monthly instalment	Reducing-balance formula
Collateral_Type	Security offered	Property / Vehicle / Gold / None

Column	Description	Range / Values
LTV_Pct	Loan-to-value ratio	Blank for unsecured loans
Co_Applicant	Co-applicant on the loan	Yes / No
Credit_Score	Bureau score	300 – 900
Credit_History_Years	Length of credit history	0.5 – 30
Existing_Loans	Active loans before this one	0 – 6
Existing_Monthly_EMI_INR	EMI on existing loans	0 if none
Credit_Utilization_Pct	Share of revolving credit used	2 – 100
Past_Delinquencies_24M	30+ day late payments, last 24 months	0 – 6
DTI_Ratio_Pct	(Existing + new EMI) / monthly income	Debt burden (FOIR)
Loan_to_Income_Ratio	Loan amount / annual income	—
Default_Flag	Target: 1 = defaulted (90+ days past due)	0 / 1

Known structural gap: `LTV_Pct` and `Collateral_Type` are null for 1,022 rows (51.1%) — this is by design, since unsecured products (Personal, Education, Business, and most Gold loans) carry no loan-to-value ratio. The notebook confirms this in Section 3: every null in these two columns corresponds exactly to a loan with no collateral, not a data quality defect.

3. Data Generation Methodology

Real loan-level data is confidential, so the dataset was generated programmatically in Python (`numpy` / `pandas`) rather than sourced from an actual lender. The goal was realism, not randomness: every field is derived from the fields before it, so the relationships a risk model would expect to find are actually present in the data.

Build order:

1. **Geography** — a state is drawn with population-weighted probability across 15 states; a city and city-tier (Metro/Tier-2/Tier-3/Rural) are drawn conditional on that state.
2. **Demographics** — age from a normal distribution (mean 36); marital status and dependents conditional on age; education level; employment type conditional on age and education (e.g., borrowers under 60 are rarely "Retired").
3. **Income** — annual income modelled as a log-normal draw whose median depends on employment type, city tier, and education level, scaled further by years in employment.
4. **Credit profile** — a latent "credit stability" score combines income, employment type, and age into a bureau score (300–900); credit history length, delinquency count, and utilization are then derived so that borrowers with a lower score also show worse delinquency and utilization patterns, as in a real bureau file.
5. **Loan terms** — loan type is drawn with type-appropriate odds (e.g., a 25-year-old is unlikely to get a Home Loan), then loan amount, tenure, and interest rate are computed together: interest rate rises as credit score falls, and loan amount is capped against a simulated affordability limit (FOIR) so EMI stays realistic relative to income. EMI itself uses the standard reducing-balance formula.
6. **Default flag** — the target variable comes from a logistic risk model, not a coin flip: the log-odds of default is a weighted sum of credit score, DTI, utilization, delinquencies, loan type, employment type, city tier, age, and loan-to-income ratio, calibrated so the overall portfolio default rate lands near realistic lending benchmarks (~12–13%).

This generation approach is why the dashboard's findings replicate well-known lending patterns — Business Loans and Gig employment carrying more risk, credit score being the dominant single predictor — even though no row of the dataset was copied from a real source.

4. Data Cleaning & Preparation

Before any feature engineering, the raw dataset was run through a validation pass in the notebook (`Untitled.ipynb`) to confirm it was analysis-ready:

Check	Method	Result
Missing values	<code>df.isnull().sum()</code>	Only <code>LTV_Pct</code> and <code>Collateral_Type</code> , 1,022 rows each — confirmed structural (unsecured loans), not a defect
Duplicate rows	<code>df.duplicated().sum()</code>	0 duplicates
Column name hygiene	<code>.str.strip().str.replace(" ", "_")</code>	No change needed — columns were already clean
Date parsing	<code>pd.to_datetime(Application_Date)</code>	0 parse failures
Text field whitespace	<code>.str.strip()</code> on all object columns	No change needed — category labels were already consistent
Target integrity	<code>Default_Flag.unique()</code>	Exactly <code>{0, 1}</code> , 0 nulls

The dataset passed validation cleanly, which confirmed it was ready for feature engineering rather than error correction. (Separately, a deliberately "dirty" version of this dataset — with duplicates, inconsistent labels, currency-as-text, and out-of-range values injected — was used as a standalone data-cleaning practice exercise; that file is not part of this pipeline.)

Feature engineering. Six fields were then derived to support dashboard filtering and segmentation, taking the working file from 31 to 37 columns:

New column	Derived from	Logic
Age_Group	Age	Binned into 6 bands: Under 25, 25-34, 35-44, 45-54, 55-64, 65+
Application_Year	Application_Date	Calendar year extracted (2022-2025)
Loan_Security	Collateral_Type	"Secured" if collateral present, else "Unsecured"
Credit_Score_Band	Credit_Score	Binned into Poor / Fair / Good / Very Good / Excellent
DTI_Band	DTI_Ratio_Pct	Binned into Low / Moderate / High / Very High / Extreme
LTV_Band	LTV_Pct	Binned into Low / Moderate / High / Very High

The enriched file was exported as `loan_default_risk_cleaned.xlsx` (2,000 rows × 37 columns) and used as the single source for the Power BI dashboard.

5. Exploratory Data Analysis (Python)

The notebook (`pandas` , `numpy` , with `matplotlib / seaborn` imported for later charting) runs a structured pass over the data before it reaches Power BI: load, profile, validate, and engineer — 40 cells in total.

Profiling. `df.shape` confirmed 2,000 rows × 31 columns; `df.head(10)` and `df.columns` were used to inspect the schema; `df.dtypes` confirmed each field's type after cleaning (dates as `datetime64` , categoricals as `str` , numerics as `int64 / float64`).

Distribution checks. Value counts were pulled for every categorical field to confirm realistic, non-degenerate splits:

Field	Top categories (share of 2,000)
Loan_Type	Personal Loan 610, Home Loan 402, Auto Loan 271, Business Loan 236, Gold Loan 209, Education Loan 176, Loan Against Property 96
Employment_Type	Salaried-Private 974, Self-Employed Business 468, Salaried-Government 213, Gig/Freelance 200, Self-Employed Professional 136, Retired 9

Field	Top categories (share of 2,000)
City_Tier	Tier-2 923, Metro 548, Tier-3 353, Rural 176
Education_Level	Graduate 899, Postgraduate 452, Higher Secondary 347, Professional 170, Below 10th 132

Target distribution. `Default_Flag.value_counts(normalize=True)` confirmed the headline split: 1,747 non-defaults (87.35%) vs. 253 defaults (12.65%) — a realistic class imbalance for a loan book, which the Power BI dashboard’s donut chart reproduces exactly.

This EDA pass intentionally stayed descriptive — profiling and validating the data — rather than producing the risk-driver breakdowns; those cross-tabulations (default rate by loan type, by credit score band, by employment type, etc.) were computed directly for this report and are reproduced as charts in Section 7 below, and power the Power BI visuals in Section 6.

6. Dashboard Design Philosophy

The dashboard was built as a single-screen, dark-themed Power BI report, with filtering concentrated in a left-hand sidebar so the full width of the canvas stays available for charts — the dashboard answers “where is risk concentrated” without the user ever scrolling.

Visual theme. A deep navy/charcoal base with a restrained gold accent (dividers, panel headers, highlighted figures) and a secondary teal accent on a handful of data points, rather than a generic light theme with primary-color charts. Every panel shares the same glass-panel treatment — soft elevation shadow, thin border, top highlight — so the report reads as one coherent surface rather than a grid of disconnected widgets.

Layout structure:

Zone	Contents	Purpose
Left sidebar	Application Year, Employment Type, Loan Type, City Tier slicers + Reset button	Global filtering, always visible

Zone	Contents	Purpose
KPI strip	Total Loans, Defaulted Loans, Default Rate, Avg Loan Amount, Avg Credit Score	Portfolio-at-a-glance
Dynamic risk chart	Horizontal bar chart + "Analyze By" field parameter	Lets the user re-cut default rate by any risk factor without switching pages
Trend & split row	Default Rate Trend (line, 2022–2025) + Defaulted vs Non-Defaulted (donut)	Time dimension + class balance
Secondary KPI row	Avg DTI, Avg LTV, Avg Interest Rate, Avg Credit Utilization, Avg Past Delinquencies	Financial and credit risk ratios
Credit score row	Default Rate by Credit Score band (column chart) + Overall Key Insights panel	Bureau-score risk gradient + narrative takeaways

DAX approach. The "Analyze By" control is a **field parameter** — a single DAX table ({("Loan Type", NAMEOF('Loan Data'[Loan_Type])), 0), ("Employment Type", NAMEOF('Loan Data'[Employment_Type])), 1), ...}) bound to one bar chart, so a single visual can be re-driven by loan type, employment type, city tier, or credit score band on demand, rather than building a separate chart per factor. Default Rate itself is a single reusable measure — `Default Rate = DIVIDE([Defaulted Loans], [Total Loans], 0)` — referenced by every KPI card and chart in the report, so the definition of "default rate" never drifts between visuals.

7. Dashboard Walkthrough

The finished Power BI report (Figure 1) puts every risk lens a credit analyst needs on one screen, filterable from the sidebar on the left.

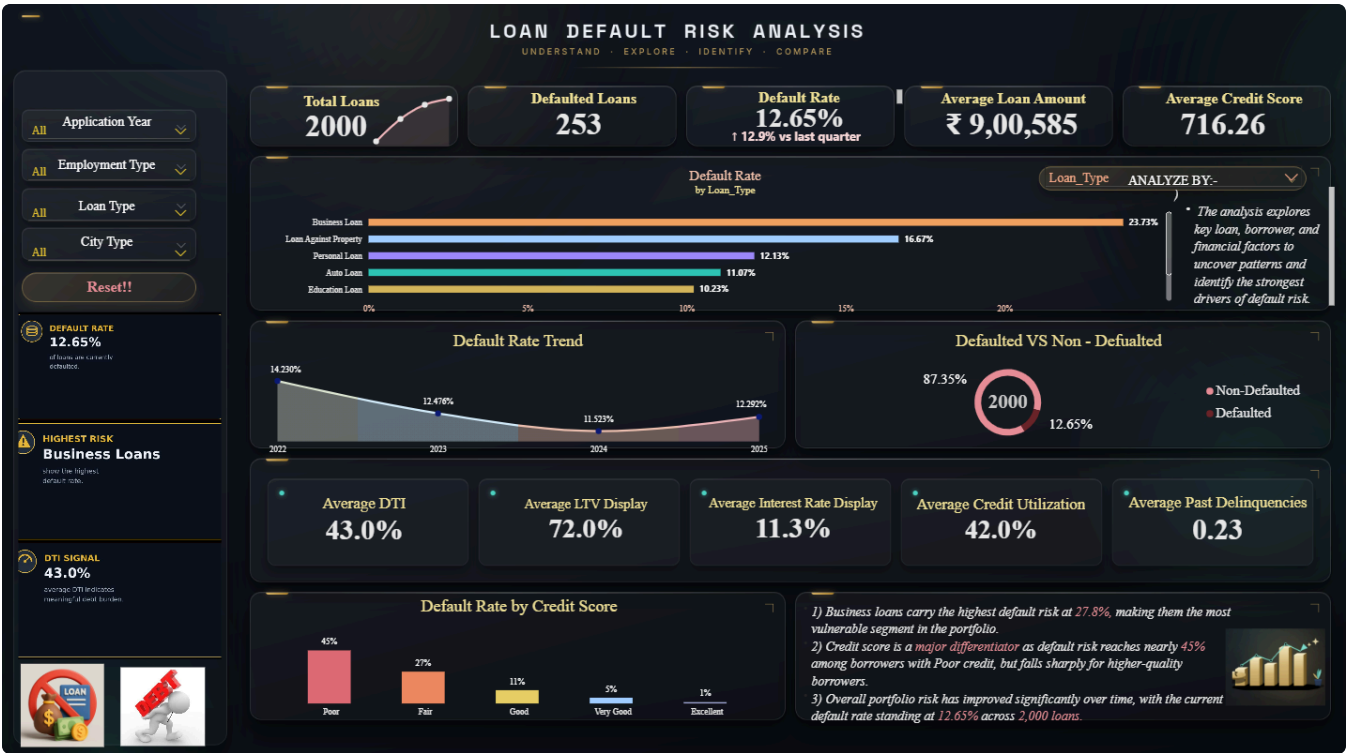


Figure 1 — the completed Loan Default Risk Analysis dashboard.

Portfolio KPIs

KPI	Value
Total Loans	2,000
Defaulted Loans	253
Default Rate	12.65%
Average Loan Amount	₹9,00,585
Average Credit Score	716.26

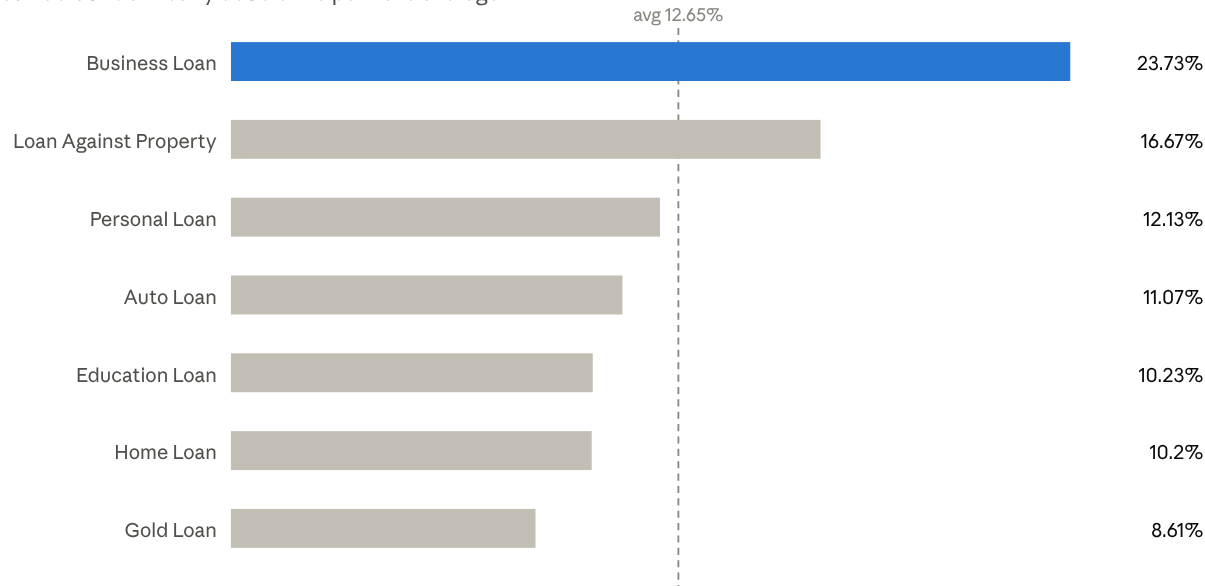
These five cards anchor the report: every other visual either breaks this 12.65% down by a risk factor or explains what sits behind it.

Default Rate by Loan Type

The dynamic risk chart — driven by the "Analyze By" field parameter — is set here to Loan Type. Business Loans stand out immediately.

Default rate by loan type

Business Loan defaults at nearly double the portfolio average



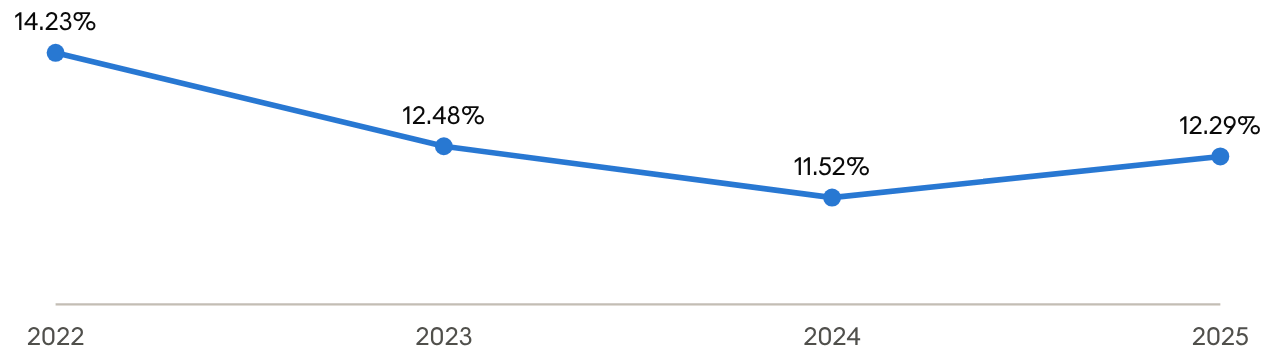
Computed from loan_default_risk_cleaned.xlsx · 2,000 loans, 7 loan types

Default Rate Trend (2022–2025)

The trend panel tracks default rate by application year: risk eased from 14.2% in 2022 to 11.5% in 2024, troughing before a modest uptick in 2025.

Default rate trend, 2022–2025

Risk eased through 2024 before ticking back up in 2025



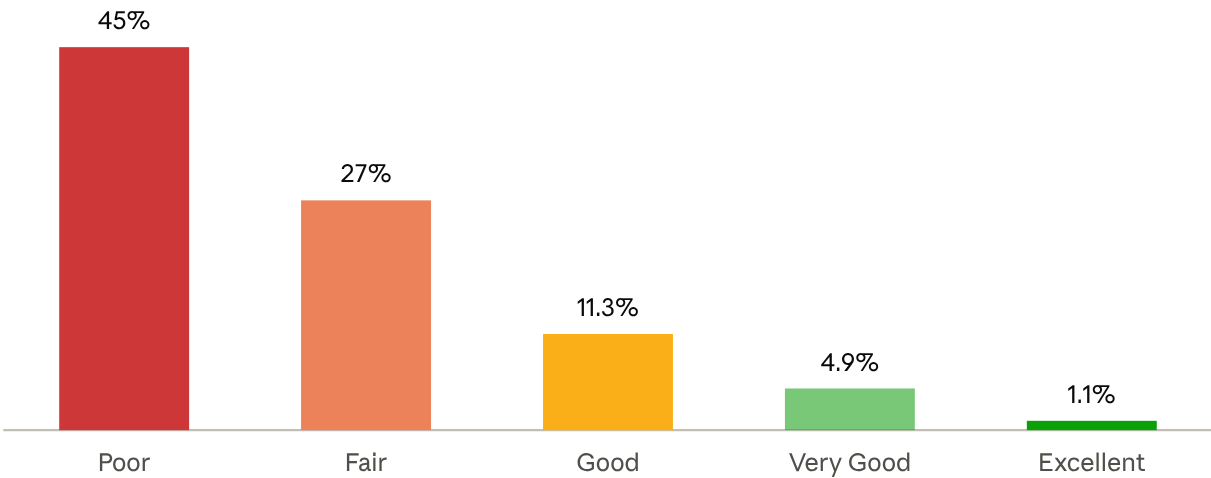
Computed from loan_default_risk_cleaned.xlsx · default rate by application year, 2022–2025

Default Rate by Credit Score Band

Bureau score is the single strongest predictor in the dataset: default rate falls from 45.0% in the Poor band to just 1.1% in the Excellent band — a 40+ point swing across five bands.

Default rate by credit score band

Risk falls sharply as bureau score improves



Computed from loan_default_risk_cleaned.xlsx · default rate by Credit_Score_Band

Secondary Risk Ratios

Metric	Portfolio Average
Debt-to-Income (DTI)	43.0%
Loan-to-Value (LTV)	72.0%
Interest Rate	11.3%
Credit Utilization	42.0%
Past Delinquencies (24M)	0.23

Sidebar Risk Signals & Insights Panel

The left sidebar surfaces three always-visible signal cards — current default rate, the highest-risk segment (Business Loans), and the average DTI — so a user sees the headline risk picture before applying any filter. The bottom-right Overall Key Insights panel translates the charts into three plain-language takeaways, detailed in Section 8 below.

8. Key Insights & Business Recommendations

1. Business Loans are the portfolio's highest-risk product (23.7% default rate — 1.9x the portfolio average). Loan Against Property is the only other product above average (16.7%); every consumer-purpose product (Home, Auto, Education, Gold) sits at or below 11%. *Recommendation: tighten underwriting criteria specifically for Business Loans — stronger collateral requirements, lower loan-to-income caps, or mandatory co-applicants — rather than applying a blanket policy change across the book.*

2. Credit score is the strongest single predictor of default. The gradient from Poor (45.0%) to Excellent (1.1%) is steep and monotonic — each band step down roughly halves or better the default rate. DTI (43.0% average) and credit utilization (42.0% average) move with it, confirming that over-leveraged, high-utilization borrowers cluster in the weaker score bands rather than appearing uniformly across the book. *Recommendation: treat a minimum credit score threshold as the single highest-leverage underwriting lever available — it outperforms any other segmentation cut in this dataset.*

3. Employment stability compounds loan-type risk. Gig/Freelance borrowers default at 27.5% and Self-Employed Business borrowers at 18.8%, versus 3.8% for Government-salaried and 5.1% for Self-Employed Professional borrowers — a nearly 7x spread across employment categories alone. Since Business Loans and self-employed/gig borrowers overlap heavily, this is very likely the mechanism behind Insight 1, not a separate, independent risk factor. *Recommendation: when reviewing a Business Loan application, weight the applicant's income stability (years in employment, employment type) at least as heavily as the loan purpose itself.*

4. Portfolio risk has genuinely improved, not just fluctuated. Default rate fell from 14.2% (2022) to a low of 11.5% (2024) before ticking back up to 12.3% in 2025 — still well below the 2022 starting point. *Recommendation: investigate what changed in the 2025 underwriting mix (loan type, geography, or score distribution) that reversed two years of improvement, before it compounds further.*

5. Secondary signal — geography. Rural borrowers default at 20.5%, nearly double the Metro rate (10.9%), and unsecured loans default at 14.5% versus 10.7% for secured loans. Both are directionally consistent with the headline insights (Rural correlates with Business/self-employed income patterns; unsecured correlates with Personal/Business loan types) rather than independent risk drivers, but are worth monitoring as the portfolio grows into new geographies.

9. Tools & Technology Stack

Stage	Tool	Used for
Data generation	Python (<code>numpy</code> , <code>pandas</code>)	Building the 2,000-row synthetic loan dataset with realistic multivariate relationships
Data storage	Excel (<code>.xlsx</code>)	Source file format for both the raw and cleaned datasets
Data validation & feature engineering	Python (<code>pandas</code>), Jupyter Notebook	Null/duplicate checks, dtype fixes, and deriving <code>Age_Group</code> , <code>Credit_Score_Band</code> , <code>DTI_Band</code> , <code>LTV_Band</code> , <code>Loan_Security</code> , <code>Application_Year</code>
Charting (exploratory)	<code>matplotlib</code> , <code>seaborn</code>	Imported in the notebook for ad hoc profiling
Dashboard	Power BI Desktop	Interactive report build
Dashboard logic	DAX	Field parameters (dynamic "Analyze By" chart) and core measures (Default Rate, averages)
Visual design	Custom dark theme (navy/gold/teal)	Report-level styling for a premium, single-screen layout

10. Challenges & Learnings

Making synthetic data behave like real data. The hardest part of this project wasn't the dashboard — it was making sure the dataset's relationships were internally consistent before any chart was built. A naive random dataset would show no real pattern between, say, credit score and default; this one needed income, credit score, delinquency history, and the default flag itself to be generated from a shared underlying model, so that the dashboard's findings would hold up to scrutiny rather than being coincidental noise.

Structural nulls vs. data quality issues. `LTV_Pct` and `Collateral_Type` are null for every unsecured loan by design — the notebook's validation step had to confirm this was expected (by checking that every null in `LTV_Pct` lines up with a null in `Collateral_Type`) rather than treating 51% missing data as a defect to fix.

Keeping one definition of "default rate" everywhere. With KPI cards, three charts, and a trend line all showing default rate, it was important to define it once as a single DAX measure and reference it everywhere, rather than letting each visual compute its own version — a mismatch between, say, a KPI card and a chart is one of the fastest ways to lose a stakeholder's trust in a dashboard.

Balancing density against readability. A risk dashboard genuinely needs a lot of numbers on screen — five KPIs, three breakdown charts, five secondary ratios, plus filters — and the main design challenge was fitting all of it on one screen without it turning into a wall of numbers. The field-parameter "Analyze By" control was the direct answer to this: instead of adding a fourth or fifth static bar chart, one chart was made to do the work of several.

What I'd do differently next time. Add a drill-through page so a specific flagged loan (from the "Highest-risk loans" view) can be opened for full detail, and validate the field-parameter chart against a couple of edge-case filter combinations (e.g., a single City Tier with very few loans) to confirm it degrades gracefully rather than showing a near-empty chart.

11. Conclusion & Future Scope

This project carries a loan portfolio end to end — from a behaviourally realistic synthetic dataset, through Python-based validation and feature engineering, to a Power BI dashboard built for how a credit risk team actually works: filter first, then drill into whichever factor matters. The headline finding is consistent across every lens in the dashboard: risk in this portfolio is concentrated, not evenly spread — in Business Loans, in Gig/self-employed income, and above all in the Poor and Fair credit score bands. A lender acting on just the top two findings (score-based underwriting thresholds and tighter Business Loan criteria) would be addressing the large majority of the portfolio's default risk.

Future scope:

- **Predictive scoring model.** Train a classification model (logistic regression or gradient boosting) on this dataset to produce a predicted default probability per loan, rather than only historical/descriptive risk rates — and surface that score directly in the dashboard.
- **Drill-through page.** Add a per-loan detail page so an analyst can click any point in the "Highest-risk loans" table and see the full borrower and loan profile.

- **Live data refresh.** Replace the static Excel source with a scheduled refresh against a database, so the dashboard reflects the portfolio in near real time.
- **Cohort and vintage analysis.** Track default rate by loan origination cohort over its lifetime (months-on-book), rather than only by calendar year, to separate seasoning effects from genuine portfolio-quality changes.
- **Early-warning alerting.** Flag borrowers whose utilization or delinquency count crosses a threshold post-disbursal, extending the dashboard from an underwriting tool into an ongoing portfolio-monitoring one.

12. Appendix

Derived fields (engineered in the notebook, not in the raw dataset):

Field	Values
Age_Group	Under 25, 25-34, 35-44, 45-54, 55-64, 65+
Application_Year	2022, 2023, 2024, 2025
Loan_Security	Secured, Unsecured
Credit_Score_Band	Poor, Fair, Good, Very Good, Excellent
DTI_Band	Low, Moderate, High, Very High, Extreme
LTV_Band	Low, Moderate, High, Very High

Full field-by-field detail for the original 31 columns is in Section 2.

Project files:

File	Contents
loan_default_risk_dataset.xlsx	Raw synthetic dataset, 2,000 rows × 31 columns, plus a Data Dictionary sheet
loan_default_risk_cleaned.xlsx	Validated + feature-engineered dataset, 2,000 rows × 37 columns
Untitled.ipynb	Python notebook — validation and feature engineering (40 cells)

File	Contents
Loan Default Risk Analysis.pbix	Power BI dashboard (not uploaded to this report; see Section 6–7 for design and walkthrough)